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Employee Attrition Prediction using Machine Learning

Project Summary

The objective of this project was to build a machine learning system capable of predicting whether an employee is likely to leave the company based on factors such as job role, salary, work-life balance, years at the company, and other HR-related attributes. Employee attrition is a major concern for organizations because losing experienced employees increases recruitment costs, training expenses, and productivity loss.

The IBM HR Analytics Employee Attrition dataset containing 1,470 employee records was used for this analysis. During the data exploration phase, it was observed that 237 employees had left the company while 1,233 employees stayed, resulting in an attrition rate of approximately 16.12%. This indicated that the dataset was imbalanced, as the majority of employees belonged to the non-attrition class.

Data preprocessing involved removing irrelevant columns such as EmployeeNumber, EmployeeCount, Over18, and StandardHours. The target variable Attrition was converted from Yes/No format into binary values (1/0). Categorical features were transformed using One-Hot Encoding, and numerical features were standardized using StandardScaler to improve model performance.

To predict employee attrition, three machine learning models were trained and compared: Logistic Regression, Random Forest Classifier, and Gradient Boosting Classifier. The models were evaluated using Precision, Recall, F1-Score, Accuracy, and ROC-AUC Score.

Among the three models, Gradient Boosting achieved the highest overall accuracy of approximately 85%. It also produced the best balance between identifying employees who might leave and minimizing false predictions. Logistic Regression achieved the highest recall score for attrition cases (62%), meaning it was the most effective at identifying employees who were likely to leave. However, its overall accuracy was lower. Random Forest achieved good accuracy but struggled to identify employees who actually left the company, resulting in very low recall for the attrition class.

The exploratory data analysis revealed several important patterns. Employees with lower monthly incomes appeared more likely to leave the organization. Attrition was also more common among employees with fewer years at the company, suggesting that employee retention is particularly important during the early stages of employment. Work-life balance showed a visible relationship with attrition, where employees reporting lower work-life balance ratings tended to leave more frequently. Certain departments and job roles also experienced higher attrition rates compared to others.

The model evaluation demonstrated that employee attrition cannot be explained by salary alone. While compensation plays an important role, other factors such as job satisfaction, career growth opportunities, work-life balance, overtime requirements, and tenure within the company also contribute significantly to an employee's decision to leave.

Based on the findings, HR teams should focus their retention efforts on employees in higher-risk roles, employees with lower satisfaction levels, and employees who are relatively new to the organization. Conducting regular employee feedback sessions, improving career development opportunities, and promoting better work-life balance initiatives may help reduce attrition rates.

One important limitation of this model is that it relies solely on historical employee data. Human decisions are influenced by many external factors such as personal circumstances, market opportunities, economic conditions, and organizational culture, which may not be fully captured in the dataset. Therefore, the model should be used as a decision-support tool rather than a replacement for human judgment.

Overall, this project successfully demonstrated how machine learning can be applied to human resource analytics to identify employees who may be at risk of leaving and provide actionable insights that support employee retention strategies.

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